T3. Frequency measurements with ring resonators — Solution

A1. They are equal to 0 because the field cannot tend to infinity at a large distance from the fiber.

$$\boxed{C_1 = 0, \qquad C_2 = 0}$$

A2. Substituting the form of the solution into the wave equation:

$$\frac{\partial^2 A}{\partial x^2} + \frac{\partial^2 A}{\partial y^2} + \left(\frac{\omega^2 \varepsilon}{c^2} - \beta^2\right) A = 0.$$

Considering that $\partial^2 A/\partial y^2 \ll \partial^2 A/\partial x^2$, and that outside the fiber $\varepsilon = 1$, we get:

$$\boxed{k_{in} = \sqrt{\frac{\omega^2 \varepsilon(\omega)}{c^2} - \beta^2}, \qquad k_{out} = \sqrt{\beta^2 - \frac{\omega^2}{c^2}}}$$

A3. We can write Faraday's law of electromagnetic induction for a rectangular circuit with vertices at the points $(a/2 - \delta a, y, z)$, $(a/2 - \delta a, y, z + \Delta z)$, $(a/2 + \delta a, y, z + \Delta z)$, $(a/2 + \delta a, y, z)$. At $\delta a \to 0$ the derivative of the magnetic flux tends to 0, while the circulation of the electric field strength keeps the same form; hence the circulation is also equal to 0. At $\delta a \to 0$ the circulation can be expressed as $\Gamma = (E_{\tau 1} - E_{\tau 2})\Delta z = 0$, hence

$$\boxed{E_{\tau 1} = E_{\tau 2}}$$

A4.

$$\boxed{B = \exp(k_{out} a/2)\cos(k_{in} a/2)}$$

A5. The proof is similar to A3, but instead of the law of electromagnetic induction, the circulation theorem for the magnetic field strength is used.

A6. Using the differential form of Faraday's law we get:

$$-\mu_0 \frac{\partial H_z}{\partial t} = \frac{\partial E_y}{\partial x}.$$

The derivative of H_z with respect to time can be expressed as

$$\frac{\partial H_z}{\partial t} = i\omega H_z,$$

and therefore

$$\boxed{H_z = E_0 \,\mathrm{Re}\left(\frac{i}{\mu_0 \omega}\frac{\partial A}{\partial x}\exp(i\omega t - i\beta z)\right)}$$

A7. To find H_z, differentiate A with respect to x:

$$\frac{\partial A}{\partial x} = -k_{in}\sin(k_{in}x), \quad |x| < a/2; \qquad \frac{\partial A}{\partial x} = -k_{out}\exp\bigl(-k_{out}(x-a/2)\bigr)\cos(k_{in}a/2), \quad x > a/2.$$

Equating H_z on both sides of the interface:

$$\boxed{\tan(k_{in}a/2) = \frac{k_{out}}{k_{in}}}$$

A8.

$$\boxed{\tan\left(\frac{a}{2}\sqrt{\frac{\omega^2\varepsilon(\omega)}{c^2} - \beta^2}\right) = \sqrt{\frac{\beta^2c^2 - \omega^2}{\omega^2\varepsilon(\omega) - \beta^2c^2}}}$$

B1.

$$\boxed{r_s^2 + t_s^2 = 1}$$

B2.

$$\boxed{B_{in}(t) = \kappa B_{out}(t - \tau(\omega))}$$

B3. Let $\varphi = -\omega\tau$. Then

$$B_{out}(t) = it_sA_{in}(t) - r_sB_{in}(t), \qquad B_{in}(t) = \kappa e^{i\varphi}B_{out}(t),$$

$$\boxed{B_{in}(t) = \kappa e^{-i\omega\tau}B_{out}(t)}$$

B4. Substituting $B_{out}(t)$ from the previous task:

$$B_{in}(t) = \kappa e^{i\varphi}\bigl(it_sA_{in}(t) - r_sB_{in}(t)\bigr), \qquad B_{in}(t) = \frac{it_s\kappa e^{i\varphi}A_{in}(t)}{1 + r_s\kappa e^{i\varphi}},$$

$$\boxed{B_{in}(t) = \frac{it_s\kappa e^{-i\omega\tau}A_0e^{i\omega t}}{1 + r_s\kappa e^{-i\omega\tau}}}$$

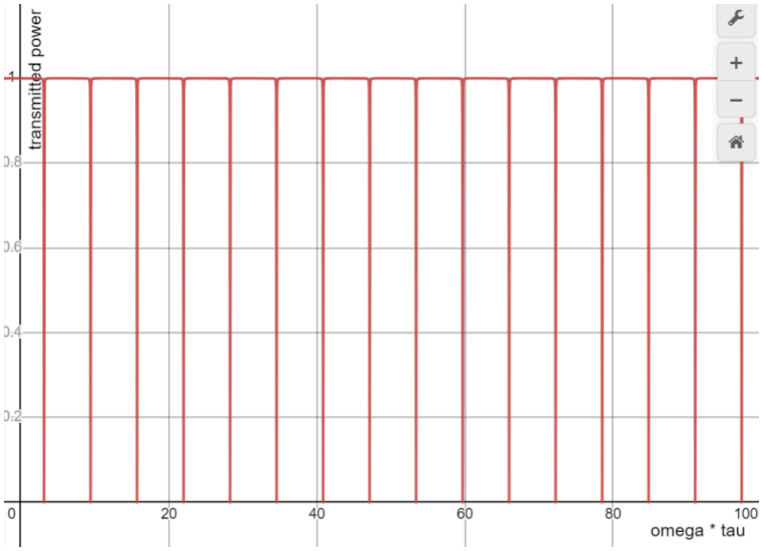
B5. The wave leaving P_2 is

$$A_{out}(t) = -r_sA_{in}(t) + it_sB_{in}(t) = -A_{in}(t)\left(r_s + \frac{t_s^2\kappa e^{i\varphi}}{1 + r_s\kappa e^{i\varphi}}\right).$$

Let us bring the expression in brackets to a common denominator and divide the squared modulus of the numerator by the squared modulus of the denominator:

$$\boxed{\eta = \frac{N_2}{N_1} = \frac{\kappa^2\sin^2(\omega\tau) + \bigl(r_s + \kappa\cos(\omega\tau)\bigr)^2}{\kappa^2r_s^2\sin^2(\omega\tau) + \bigl(1 + r_s\kappa\cos(\omega\tau)\bigr)^2}}$$

B6.



The transmitted power N_2/N_1 against $\omega\tau$.

$$\boxed{\omega_{res}\tau = \pi(1+2n)}$$

B7. Let us find the depth of the absorption peak by expanding the expression for the power ratio in a Taylor series:

$$\eta(\omega_{res}) \approx \frac{(1-\kappa-t_s^2/2)^2}{(1-\kappa+t_s^2/2)^2} \approx 0, \qquad \Delta\eta_{max} = 1-\eta(\omega_{res}) \approx 1,$$

or by substituting the value of φ from the previous item without expanding: $\eta(\omega_{res}) \approx 1.6\cdot 10^{-6} \approx 0$, $\Delta\eta_{max} \approx 1$.

The sharpness is the ratio of ω to the width of the region in which $\Delta\eta > \Delta\eta_{max}/2$. Let us estimate the width of this region. Put

$$\varphi = \delta + \pi(1+2n), \qquad \delta \ll \pi,$$

and expand in a Maclaurin series in δ:

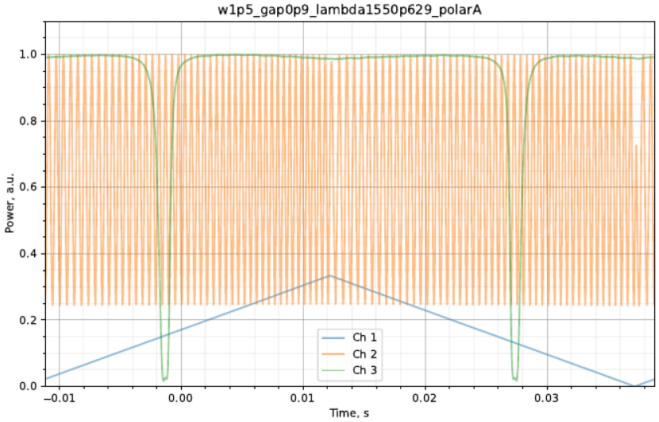
$$\eta \approx \frac{\delta^2 + (1-\kappa-t_s^2/2+\delta^2/2)^2}{\delta^2 + (1-\kappa+t_s^2/2+\delta^2/2)^2} \approx \frac{\delta^2}{\delta^2 + (1-\kappa+t_s^2/2)^2} = \frac{1}{2}.$$

From here we get

$$\delta \approx 1-\kappa+t_s^2/2 \approx 0.01,$$

$$\boxed{Q = \frac{\omega_{res}}{\Delta\omega} = \frac{\pi(1+2n)}{2\delta} \approx \frac{\pi n}{1-\kappa+t_s^2/2} = \pi\cdot 10^4 = 3.14\cdot 10^4}$$

C1.

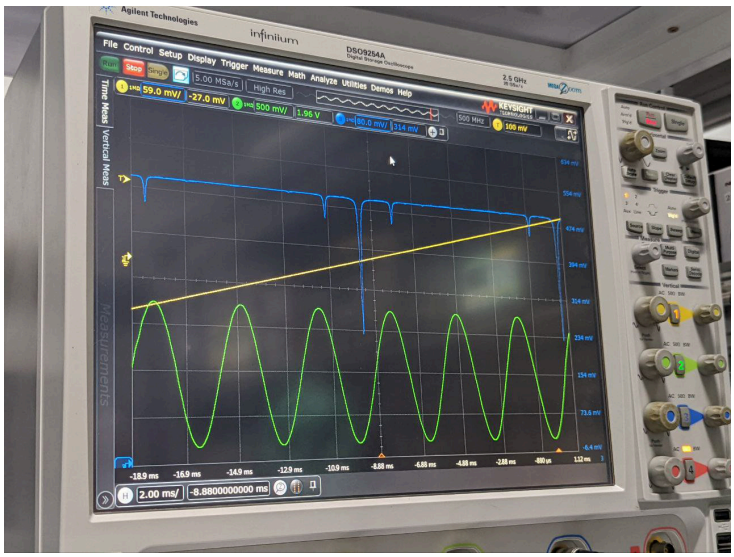


Oscilloscope readings: the interferometer signal and the resonator signal with its absorption dip.

C2. The field amplitude at the exit of the EOM equals

$$E_{EOM}(t) = f(t)\cos(\omega(t)t) = \beta\cos(\omega t) + \frac{1-\beta}{2}\Big(\cos(\Omega+\omega)t + \cos(-\Omega+\omega)t\Big),$$

where $\omega = \omega(t) = \omega_0 + \alpha t$. In this case three waves fall on the FR, with frequencies shifted relative to each other by exactly Ω. Therefore the number of absorption peaks triples: they can be observed at the frequencies ω_{res}, $\omega_{res}+\Omega$, $\omega_{res}-\Omega$, where ω_{res} is the frequency of minimum transmission (defined in B6).



What the oscilloscope shows with the modulator on.

C3. We can measure one period by changing Ω so that the absorption maxima are separated by ω_{MZI} from each other. Then the absolute error is of order $\omega_0/Q = 24$ MHz, and the relative error is

$$\boxed{\frac{\omega_0}{Q\,\Omega_{max}} = 0.02}$$

D1.

$$\boxed{v_g = 1/\beta_1}$$

D2. Let us calculate the derivatives and substitute them into the equation (dividing by $iF_0 \exp(i\sigma z)$):

$$\frac{\sigma}{\cosh \theta s} + \frac{\beta_2 \theta^2}{2\beta_1^2}\left(-\frac{2}{\cosh^3 \theta s} + \frac{1}{\cosh \theta s}\right) = \frac{\gamma F_0^2}{\cosh^3 \theta s}.$$

From here we get

$$\boxed{F_0^2 = -\frac{\beta_2 \theta^2}{\gamma \beta_1^2}, \qquad \sigma = -\frac{\beta_2 \theta^2}{2\beta_1^2}}$$

D3. Using the result of B3, the natural-frequency condition reads

$$\beta(\omega_\mu) L + \pi = 2\pi(\mu + \mu_0).$$

Now let us substitute the expression for $\beta(\omega)$, and into it ω_μ. We get

$$\left(\beta_0 + \beta_1\left(D_1\mu + \frac{D_2\mu^2}{2}\right) + \frac{\beta_2 D_1^2 \mu^2}{2}\right) L \approx 2\pi(\mu + \mu_0 + \text{const}).$$

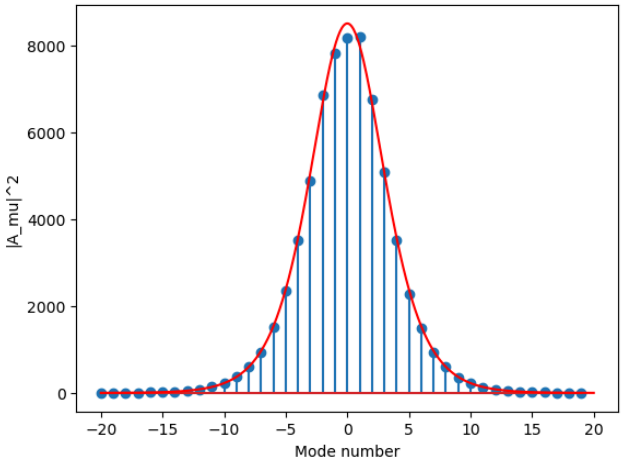
Equating the coefficients of μ and μ^2 on the two sides:

$$\boxed{D_1 = \frac{2\pi}{\beta_1 L}, \qquad D_2 = -\frac{\beta_2 D_1^2}{\beta_1} = -\frac{4\pi^2 \beta_2}{\beta_1^3 L^2}}$$

D4. $F_0^2 > 0$, so $\beta_2 < 0$ (as γ has the sign of $\chi > 0$), and then

$$\boxed{D_2 > 0}$$

D5. The spectrum consists of narrow bands at the natural frequencies of the FR, as shown in the figure. The envelope can be fitted by $F_1/\cosh^2\big(F_2(\omega-\omega_0)\big)$ (here F_1, F_2 are some constants depending on F_0, σ, θ), but an analytical expression for the envelope is not required.



Emission spectrum of the resonator: lines at the natural frequencies $\omega_0 + D_1\mu$ (horizontal axis: mode number μ).

D6. We get a ruler with a pitch of D_1, and the absolute error is about half of this pitch:

$$\boxed{\Delta\omega \approx \frac{D_1}{2}}$$

D7.

$$\boxed{\tau_s = \frac{2\pi}{D_1}}$$